

Standard Model Signals in the Cosmological Higgs Collider

1 Gauge Boson Signal

The Higgs bulk-to-boundary propagator is massless:

$$G_{\pm}(k; \tau) = \frac{H^2}{2k^3} (1 \mp ik\tau) e^{\pm ik\tau} \quad (1)$$

The Higgs–gauge couplings arise from the original SM Lagrangian:

$$\frac{\sqrt{-g}}{2} |D_{\mu} \mathbf{H}|^2 \supset \frac{a(\tau)^2}{4} g^2 (h_0 + \delta h)^2 \left(W_{\mu}^+ W^{-\mu} + \frac{1}{2c_W^2} Z^{\mu} Z_{\mu} \right) \quad (2)$$

The gauge-boson masses induced by infrared Higgs-loop corrections are:

$$m_W^2 = \frac{3g^2 H^4}{8\pi^2 m_H^2}, \quad m_Z^2 = \frac{m_W^2}{c_W^2} \quad (3)$$

The mode functions of a massive gauge boson are given by [1]:

$$B^{(\pm)}(k, \tau) = \frac{1}{\sqrt{2k}} W_{0, i\tilde{\nu}}(2ik\tau), \quad (4)$$

$$B^{(T)}(k, \tau) = \frac{\sqrt{\pi}k}{2m} e^{-\pi\tilde{\nu}/2} H(-\tau)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau), \quad (5)$$

$$B^{(L)}(k, \tau) = -\frac{i\sqrt{\pi}}{4m} e^{-\pi\tilde{\nu}/2} H(-\tau)^{1/2} \left[H_{i\tilde{\nu}}^{(1)}(-k\tau) - k\tau \left(H_{1+i\tilde{\nu}}^{(1)}(-k\tau) - H_{-1+i\tilde{\nu}}^{(1)}(-k\tau) \right) \right] \quad (6)$$

where $\tilde{\nu} = \sqrt{(m/H)^2 - 1/4}$. The Whittaker W function at zero chemical potential can be transformed into a Hankel function:

$$\begin{aligned} W_{0, i\tilde{\nu}}(2ik\tau) &= \sqrt{\frac{2ik\tau}{\pi}} K_{i\tilde{\nu}}(ik\tau) \\ &= \sqrt{\frac{i\pi k\tau}{2}} i e^{-\pi\tilde{\nu}/2} H_{i\tilde{\nu}}^{(1)}(-k\tau) \end{aligned} \quad (7)$$

Thus, the greater propagator, or $-+$ propagator, for each polarization is:

$$D_{>}^{(\pm)}(k; \tau_1, \tau_2) = B^{(\pm)}(k, \tau_1) B^{(\pm)*}(k, \tau_2) = \frac{\pi e^{-\pi\tilde{\nu}} (\tau_1 \tau_2)^{1/2}}{4} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2), \quad (8)$$

$$D_{>}^{(T)}(k; \tau_1, \tau_2) = B^{(T)}(k, \tau_1) B^{(T)*}(k, \tau_2) = \frac{\pi H^2 k^2}{4m^2} e^{-\pi\tilde{\nu}} (\tau_1 \tau_2)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2), \quad (9)$$

$$D_{>}^{(L)}(k; \tau_1, \tau_2) = \frac{1}{k^2} \left(\partial_{\tau_1} - \frac{2}{\tau_1} \right) \left(\partial_{\tau_2} - \frac{2}{\tau_2} \right) D_{>}^{(T)}(k; \tau_1, \tau_2). \quad (10)$$

For reference, the physical scalar propagator with mass index $i\tilde{\nu}$ is:

$$D_{>}^{(\text{scalar})} = \frac{\pi H^2}{4} e^{-\pi\tilde{\nu}} (\tau_1 \tau_2)^{3/2} H_{i\tilde{\nu}}^{(1)}(-k\tau_1) H_{-i\tilde{\nu}}^{(2)}(-k\tau_2) \quad (11)$$

and the nonlocal part of its late-time expansion is:

$$[D_{>}^{(\text{scalar})}]_{\text{NonL}} = [D_{<}^{(\text{scalar})}]_{\text{NonL}} = \frac{H^2(\tau_1 \tau_2)^{3/2}}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (12)$$

The vector propagators can be related to the scalar propagator as follows:

$$D_{>}^{(\pm)} = \frac{1}{H^2 \tau_1 \tau_2} D_{>}^{(\text{scalar})} \quad (13)$$

$$D_{>}^{(T)} = \frac{k^2}{H^2(\tilde{\nu}^2 + 1/4)} D_{>}^{(\text{scalar})} \quad (14)$$

$$D_{>}^{(L)} = \frac{1}{H^2(\tilde{\nu}^2 + 1/4)} \left(\partial_{\tau_1} - \frac{2}{\tau_1} \right) \left(\partial_{\tau_2} - \frac{2}{\tau_2} \right) D_{>}^{(\text{scalar})}(k; \tau_1, \tau_2) \quad (15)$$

The propagator for the temporal component of the longitudinal mode is suppressed by a factor of $(k\tau)^2$. It therefore contributes only at next-to-leading order and is neglected below.

The corresponding nonlocal parts are:

$$[D^{(\pm)}]_{\text{NonL}} = \frac{(\tau_1 \tau_2)^{1/2}}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (16)$$

$$[D^{(T)}]_{\text{NonL}} = \frac{k^2}{\tilde{\nu}^2 + 1/4} \frac{(\tau_1 \tau_2)^{3/2}}{4\pi} \left[\Gamma^2(-i\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (17)$$

$$[D^{(L)}]_{\text{NonL}} = \frac{(\tau_1 \tau_2)^{1/2}}{4\pi(\tilde{\nu}^2 + 1/4)} \left[(-1/2 + i\tilde{\nu})^2 \Gamma^2(-i\tilde{\nu}) \left(\frac{k^2 \tau_1 \tau_2}{4} \right)^{i\tilde{\nu}} + \text{c.c.} \right] \quad (18)$$

We also define the propagators for the helicity eigenstates:

$$D_{>\mu\nu}^{(\pm 1)}(\mathbf{k}; \tau_1, \tau_2) = e_{\mu, \mathbf{k}}^{(\pm)} e_{\nu, \mathbf{k}}^{(\pm)*} D_{>}^{\pm}(k; \tau_1, \tau_2) \quad (19)$$

$$D_{>\mu\nu}^{(0)}(\mathbf{k}; \tau_1, \tau_2) = e_{\mu, \mathbf{k}}^{(L)} e_{\nu, \mathbf{k}}^{(L)*} D_{>}^{(L)}(k; \tau_1, \tau_2) \quad (20)$$

$$D_{>\mu\nu}(\mathbf{k}; \tau_1, \tau_2) = \sum_{h=\pm 1, 0} D_{>\mu\nu}^{(h)}(k; \tau_1, \tau_2) \quad (21)$$

For $\mathbf{k} = (0, 0, k)$, the polarization vectors are:

$$e_{\mu, \mathbf{k}}^{(T)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad e_{\mu, \mathbf{k}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix}, \quad e_{\mu, \mathbf{k}}^{(L)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad k_{\mu} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ k \end{pmatrix}, \quad (22)$$

When we rotate $\mathbf{k}$, $e_{\mu, \mathbf{k}}^{(T)}$ remains invariant, while the other two types of polarization vectors transform as $e'_{i, \mathbf{k}} = R_{ij} e_{j, \mathbf{k}}$:

$$e_{\mu, \mathbf{k}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \cos \theta \cos \phi \mp i \sin \phi \\ \cos \theta \sin \phi \pm i \cos \phi \\ -\sin \theta \end{pmatrix}, \quad k_{\mu} = e_{\mu, \mathbf{k}}^{(L)} = \begin{pmatrix} 0 \\ \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} \quad (23)$$

The MB representation below uses the rescaled scalar seed $D^{(\text{scalar})}/H^2$:

$$\frac{D_{\geq}^{(\text{scalar})}(k; \tau_1, \tau_2)}{H^2} = \frac{1}{4\pi} \int_{-i\infty}^{+i\infty} \frac{ds_1}{2\pi i} \frac{ds_2}{2\pi i} e^{\pm i\pi(s_1-s_2)} \left(\frac{k}{2}\right)^{-2s_{12}} (-\tau_1)^{-2s_1+3/2} (-\tau_2)^{-2s_2+3/2} \quad (24)$$

$$\times \Gamma\left[s_1 - \frac{i\tilde{\nu}}{2}, s_1 + \frac{i\tilde{\nu}}{2}, s_2 - \frac{i\tilde{\nu}}{2}, s_2 + \frac{i\tilde{\nu}}{2}\right] \quad (25)$$

For simplicity, we define the following loop seeds:

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} |\mathbf{q}|^{-2a} |\mathbf{q} + \mathbf{k}_s|^{-2b} = \frac{k_s^{3-2a-2b}}{(4\pi)^{3/2}} \cdot f_0(a, b) \quad (26)$$

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} |\mathbf{q}|^{-2a} |\mathbf{q} + \mathbf{k}_s|^{-2b} \cos\theta = \frac{k_s^{3-2a-2b}}{(4\pi)^{3/2}} \cdot f_c(a, b) \quad (27)$$

$$\int \frac{d^3\mathbf{q}}{(2\pi)^3} |\mathbf{q}|^{-2a} |\mathbf{q} + \mathbf{k}_s|^{-2b} \cos^2\theta = \frac{k_s^{3-2a-2b}}{(4\pi)^{3/2}} \cdot f_{cc}(a, b) \quad (28)$$

where $f_0(a, b)$, $f_c(a, b)$, and $f_{cc}(a, b)$ are combinations of Gamma functions arising from the loop integrals:

$$f_0(a, b) = \Gamma\left[\begin{matrix} a+b-\frac{3}{2}, \frac{3}{2}-a, \frac{3}{2}-b \\ 3-a-b, a, b \end{matrix}\right] \quad (29)$$

$$f_c(a, b) = \frac{1}{2}f_0(a+1/2, b-1) - \frac{1}{2}f_0(a-1/2, b) - \frac{1}{2}f_0(a+1/2, b) \quad (30)$$

$$\begin{aligned} f_{cc}(a, b) &= \frac{1}{4}f_0(a+1, b-2) + \frac{1}{4}f_0(a-1, b) + \frac{1}{4}f_0(a+1, b) \\ &\quad - \frac{1}{2}f_0(a, b-1) - \frac{1}{2}f_0(a+1, b-1) + \frac{1}{2}f_0(a, b) \end{aligned} \quad (31)$$

2 Bubble Diagram

For the bubble diagram, we have:

$$\begin{aligned} \langle \delta h^4 \rangle^{\text{Bubble}} &= (g^2)^2 \sum_{a,b=\pm} (-ab) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} G_a(k_1; \tau_1) G_a(k_2; \tau_1) G_b(k_3; \tau_2) G_b(k_4; \tau_2) \\ &\quad \times \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{\mu\nu}(q, \tau_1, \tau_2)]_{ab} [D^{\nu\mu}(|\mathbf{k}_s + \mathbf{q}|, \tau_2, \tau_1)]_{ba} \end{aligned} \quad (32)$$

The relevant loop integral is:

$$\mathcal{LOOP}_{ab} = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{\mu\nu}(q, \tau_1, \tau_2)]_{ab} [D^{\nu\mu}(|\mathbf{k}_s + \mathbf{q}|, \tau_2, \tau_1)]_{ba} \quad (33)$$

The cutting rule implies that the nonlocal signal is independent of the SK index:

$$\begin{aligned} \mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(L)i} e_{\mathbf{q}}^{(L)j*} + D^{(\pm)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(\pm)i} e_{\mathbf{q}}^{(\pm)j*} \right] \\ &\quad \times \left[D^{(L)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)i*} + D^{(\pm)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(\pm)j} e_{\mathbf{K}}^{(\pm)i*} \right] \end{aligned} \quad (34)$$

$$= \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1\sigma_2} \quad (35)$$

where $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$. We therefore consider the different polarization combinations. Setting $\mathbf{k}_s = k_s(0, 0, 1)$ and $\mathbf{q} = q(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, the polarization vectors become:

$$e_{i,\mathbf{q}}^{(L)} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}, \quad e_{i,\mathbf{q}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos \theta \cos \phi \mp i \sin \phi \\ \cos \theta \sin \phi \pm i \cos \phi \\ -\sin \theta \end{pmatrix} \quad (36)$$

$$\mathbf{e}_{i,\mathbf{K}}^{(L)} = \frac{1}{K} \begin{pmatrix} q \sin \theta \cos \phi \\ q \sin \theta \sin \phi \\ q \cos \theta + k_s \end{pmatrix}, \quad \mathbf{e}_{i,\mathbf{K}}^{(\pm)} = \frac{1}{\sqrt{2}K} \begin{pmatrix} \cos \phi (q \cos \theta + k_s) \mp i \sin \phi K \\ \sin \phi (q \cos \theta + k_s) \pm i \cos \phi K \\ -q \sin \theta \end{pmatrix} \quad (37)$$

The signal arises from the loop integral over the nonlocal part of each propagator. We calculate the first term as an example:

$$\begin{aligned} \mathcal{LOOP}_{LL} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(L)i} e_{\mathbf{q}}^{(L)j*} \cdot D^{(L)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)i*} \right]_{\text{NonL}} \\ &= \sum_{c=\pm} \frac{(-1/2 + ic\tilde{\nu})^4 (\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(\tilde{\nu}^2 + 1/4)^2 (4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[\left(\frac{q}{2}\right)^{2ic\tilde{\nu}} \left(\frac{K}{2}\right)^{2ic\tilde{\nu}} |e_{\mathbf{q}}^{(L)} \cdot e_{\mathbf{K}}^{(L)*}|^2 \right] \end{aligned} \quad (38)$$

Performing the loop integral gives:

$$\begin{aligned} \mathcal{LOOP}_{LL} &= \sum_{c=\pm} \frac{(-1/2 + ic\tilde{\nu})^4 (\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(\tilde{\nu}^2 + 1/4)^2 (4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \cdot \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\quad \times (f_0(x-1, x+1) + 2f_c(x-1/2, x+1) + f_{cc}(x, x+1)) \end{aligned} \quad (39)$$

where $x = -ic\tilde{\nu}$. Using the identity $\mathbf{q} \cdot \mathbf{K} = \frac{1}{2}(q^2 + K^2 - k_s^2)$, we can instead write it as:

$$\begin{aligned} \mathcal{LOOP}_{LL} &= \sum_{c=\pm} \frac{(-1/2 + ic\tilde{\nu})^4 (\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(\tilde{\nu}^2 + 1/4)^2 (4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \cdot \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\quad \times \frac{1}{4} (f_0(x-1, x+1) + f_0(x+1, x-1) + f_0(x+1, x+1) \\ &\quad + 2f_0(x, x) - 2f_0(x, x+1) - 2f_0(x+1, x)) \end{aligned} \quad (40)$$

We define it by:

$$\mathcal{LOOP}_{LL} = \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{LL}(-ic\tilde{\nu}) \quad (41)$$

For the transverse mode, we similarly have:

$$\begin{aligned} \mathcal{LOOP}_{++} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(+)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(+i)} e_{\mathbf{q}}^{(+j)*} \cdot D^{(+)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(+j)} e_{\mathbf{K}}^{(+i)*} \right]_{\text{NonL}} \\ &= \sum_{c=\pm} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[\left(\frac{q}{2}\right)^{2ic\tilde{\nu}} \left(\frac{K}{2}\right)^{2ic\tilde{\nu}} |e_{\mathbf{q}}^{(+)} \cdot e_{\mathbf{K}}^{(+)*}|^2 \right] \end{aligned} \quad (42)$$

Taking the vector structure into account gives:

$$\begin{aligned}\mathcal{LOOP}_{++} &= \sum_{c=\pm} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \times \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\times \frac{1}{4} [f_0(x, x) + f_0(x-1, x+1) + f_{cc}(x, x+1) \\ &+ 2f_0(x - \frac{1}{2}, x + \frac{1}{2}) + 2f_c(x - \frac{1}{2}, x+1) + 2f_c(x, x + \frac{1}{2})] \end{aligned} \quad (43)$$

$$= \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{++}(-ic\tilde{\nu}) \quad (44)$$

We note that the $--$ term equals the $++$ term. We next consider the mixed terms, beginning with the $L+$ term:

$$\begin{aligned}\mathcal{LOOP}_{L+} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(q, \tau_1, \tau_2) e_{\mathbf{q}}^{(L)i} e_{\mathbf{q}}^{(L)j*} \cdot D^{(+)}(K, \tau_2, \tau_1) e_{\mathbf{K}}^{(+j)} e_{\mathbf{K}}^{(+i)*} \right]_{\text{NonL}} \\ &= \sum_{c=\pm} \frac{(-1/2 + ic\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[\left(\frac{q}{2}\right)^{2ic\tilde{\nu}} \left(\frac{K}{2}\right)^{2ic\tilde{\nu}} |e_{\mathbf{q}}^{(L)} \cdot e_{\mathbf{K}}^{(+)*}|^2 \right] \end{aligned} \quad (45)$$

which gives:

$$\begin{aligned}\mathcal{LOOP}_{L+} &= \sum_{c=\pm} \frac{(-1/2 + ic\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \cdot \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\times \frac{1}{2} (f_0(x, x+1) - f_{cc}(x, x+1)) \end{aligned} \quad (46)$$

$$= \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{L+}(-ic\tilde{\nu}) \quad (47)$$

Evaluating the dot products, we find that $L- = L+ = -L = +L$.

For the $+-$ term:

$$\begin{aligned}\mathcal{LOOP}_{+-} &= \sum_{c=\pm} \frac{(\tau_1 \tau_2)^{1+2ic\tilde{\nu}}}{(4\pi)^2} \Gamma^4(-ic\tilde{\nu}) \times \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \\ &\times \frac{1}{4} [f_0(x, x) + f_0(x-1, x+1) + f_{cc}(x, x+1) \\ &- 2f_0(x - \frac{1}{2}, x + \frac{1}{2}) + 2f_c(x - \frac{1}{2}, x+1) - 2f_c(x, x + \frac{1}{2})] \end{aligned} \quad (48)$$

$$= \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \mathcal{I}_{+-}(-ic\tilde{\nu}) \quad (49)$$

Similarly, $-- = +-.$

Thus, the full loop integral is:

$$\mathcal{LOOP} = \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1 \sigma_2} = \sum_{c=\pm} (\tau_1 \tau_2)^{1+2ic\tilde{\nu}} \cdot \mathcal{B}_c \quad (50)$$

where the bubble subgraph contribution is:

$$\mathcal{B}_c = \frac{\Gamma^4(-ic\tilde{\nu})}{(4\pi)^2} \frac{4^{-2ic\tilde{\nu}} k_s^{3+4ic\tilde{\nu}}}{(4\pi)^{3/2}} \cdot \sum_{\lambda_1, \lambda_2=L, \pm} \mathcal{I}_{\lambda_1 \lambda_2}(-ic\tilde{\nu}) \quad (51)$$

We now consider the contributions from the left and right subgraphs, namely, the time integrals. For the left subgraph:

$$\begin{aligned}
\mathcal{T}_c^{(L)} &= (g^2) \sum_a ia \int \frac{d\tau_1}{|H\tau_1|^2} G_a(k_1; \tau_1) G_a(k_2; \tau_1) (-\tau_1)^{1+2ic\tilde{\nu}} \\
&= (g^2) \frac{H^2}{4k_1^3 k_2^3} \sum_{a=\pm} ia \int_{-\infty}^0 d\tau_1 (-\tau_1)^{-1+2ic\tilde{\nu}} (1 - ia k_{12} \tau_1 - k_1 k_2 \tau_1^2) e^{iak_{12}\tau_1} \\
&= (g^2) \frac{H^2}{4k_1^3 k_2^3} \times k_{12}^{-2ic\tilde{\nu}} \cdot \mathcal{P}_c^{(1)}(\tilde{\nu})
\end{aligned} \tag{52}$$

where the coefficient is:

$$\mathcal{P}_c^{(1)}(\tilde{\nu}) = \frac{2 + ic\tilde{\nu}}{2ic\tilde{\nu}} i \sinh(\pi c\tilde{\nu}) \Gamma(2 + 2ic\tilde{\nu}) \tag{53}$$

Similarly, for the right subgraph:

$$\mathcal{T}_c^{(R)} = (g^2) \frac{H^2}{4k_3^3 k_4^3} \times k_{34}^{-2ic\tilde{\nu}} \cdot \mathcal{P}_c^{(1)}(\tilde{\nu}) \tag{54}$$

Thus, the total contribution of the bubble diagram is:

$$\langle \delta h^4 \rangle^{\text{Bubble}} = \sum_{c=\pm} \mathcal{T}_c^{(L)} \mathcal{T}_c^{(R)} \mathcal{B}_c \tag{55}$$

$$= (g^2)^2 \frac{H^4 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{\text{Bub}}(\tilde{\nu}) + \text{c.c.} \tag{56}$$

where the magnitude is:

$$C^{\text{Bub}}(\tilde{\nu}) = [\mathcal{P}_+^{(1)}(\tilde{\nu})]^2 \cdot \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot \sum_{\lambda_1, \lambda_2=L, \pm} \mathcal{I}_{\lambda_1 \lambda_2}(\tilde{\nu})|_{c=+} \tag{57}$$

A numerical result is shown below:

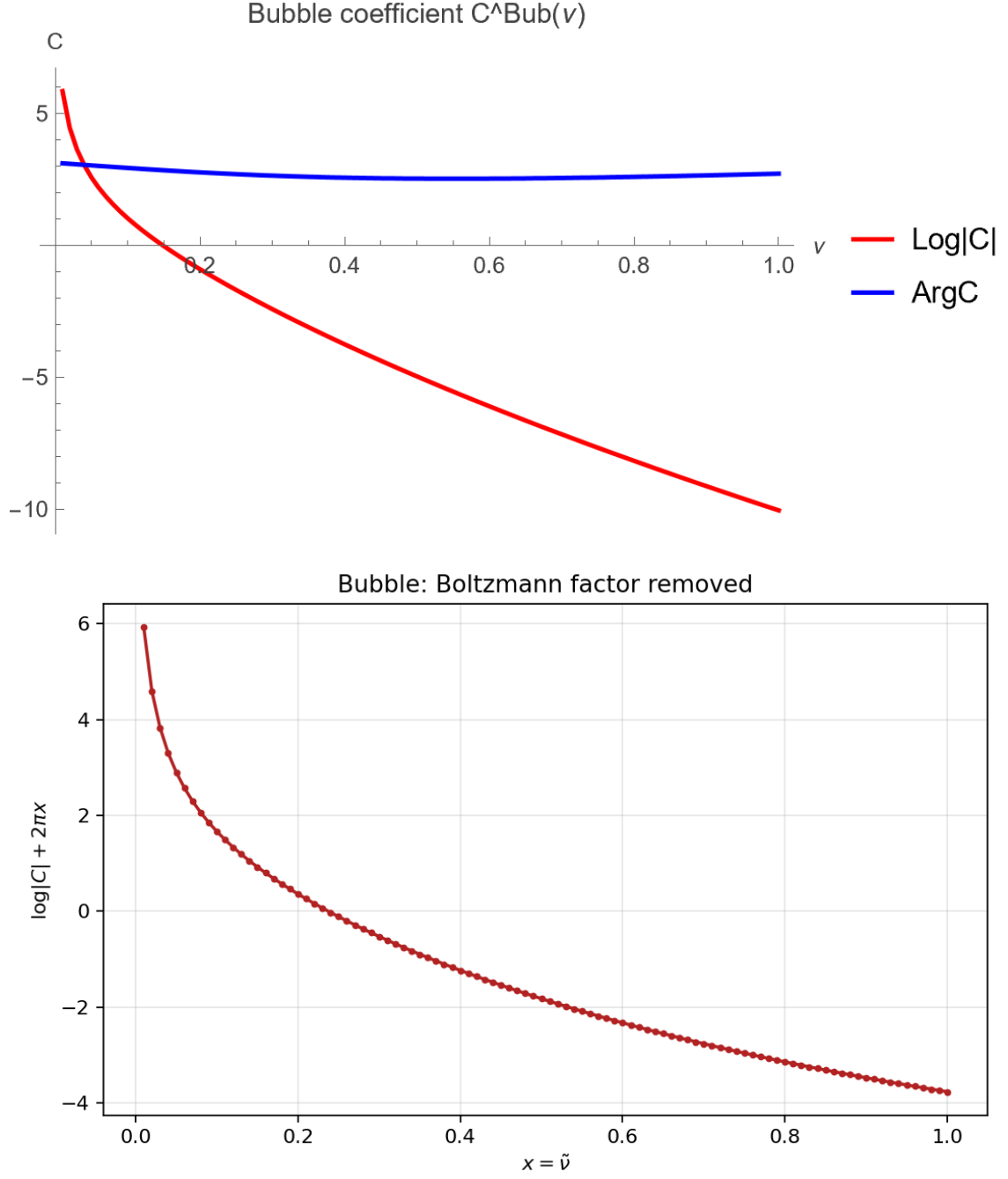


Figure 1: $C_{\text{Bub}}(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |C_{\text{Bub}}(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom), with the leading $e^{-2\pi\tilde{\nu}}$ factor divided out.

3 Triangle Diagram

One of the triangle diagrams is given by:

$$\begin{aligned} \langle \delta h^4 \rangle^{\text{Tri1}} &= (g^2) (g^2 h_0)^2 \sum_{a,b,c=\pm} (-iabc) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} \frac{d\tau_3}{|H\tau_3|^2} G_a(k_1; \tau_1) G_b(k_2; \tau_2) G_c(k_3; \tau_3) G_c(k_4; \tau_3) \\ &\times \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} [D_{ki}(\mathbf{q}, \tau_3, \tau_1)]_{ca} \end{aligned} \quad (58)$$

First, we consider the loop integral:

$$\mathcal{LOOP} = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} [D_{ki}(\mathbf{q}, \tau_3, \tau_1)]_{ca} \quad (59)$$

According to the cutting rule, this integral contributes to the nonlocal signal only when both $\mathbf{q}$ and $\mathbf{k}_s$ become soft.

$$\mathcal{LOOP} = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)] [D_{ki}(\mathbf{q}, \tau_3, \tau_1)] \quad (60)$$

We use:

$$e_{i,\mathbf{k}_1}^{(L)} = \begin{pmatrix} \sin \theta_1 \\ 0 \\ \cos \theta_1 \end{pmatrix}, \quad e_{i,\mathbf{k}_1}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos \theta_1 \\ \pm i \\ -\sin \theta_1 \end{pmatrix}. \quad (61)$$

Expanding the propagators in the helicity basis gives:

$$\begin{aligned} \mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(L)i} e_{\mathbf{k}_1}^{(L)j*} + D^{(\pm)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(\pm)i} e_{\mathbf{k}_1}^{(\pm)j*} \right]_{ab} \\ &\times \left[D^{(L)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)k*} + D^{(\pm)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(\pm)j} e_{\mathbf{K}}^{(\pm)k*} \right] \\ &\times \left[D^{(L)}(q, \tau_3, \tau_1) e_{\mathbf{q}}^{(L)k} e_{\mathbf{q}}^{(L)i*} + D^{(\pm)}(q, \tau_3, \tau_1) e_{\mathbf{q}}^{(\pm)k} e_{\mathbf{q}}^{(\pm)i*} \right] \end{aligned} \quad (62)$$

$$= \sum_{\lambda, \sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1 \sigma_2}^\lambda \quad (63)$$

where the contribution from each polarization combination is:

$$\begin{aligned} \mathcal{LOOP}_{\sigma_1 \sigma_2}^\lambda &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D^{(\lambda)}(k_1, \tau_1, \tau_2)]_{ab} D^{(\sigma_1)}(K, \tau_2, \tau_3) D^{(\sigma_2)}(q, \tau_3, \tau_1) \\ &\times e_{\mathbf{k}_1}^{(\lambda)i} e_{\mathbf{k}_1}^{(\lambda)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{q}}^{(\sigma_2)k} e_{\mathbf{q}}^{(\sigma_2)i*} \end{aligned} \quad (64)$$

First, we consider the $\lambda = L$ terms and evaluate the contractions with the other two propagators. We set $\mathbf{k}_1 = k_1(\sin \theta_1, 0, \cos \theta_1)$, and still use $\mathbf{k}_s = k_s(0, 0, 1)$ and $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$. Then the vector structure is:

$$\begin{aligned} \mathcal{E}_{\sigma_1 \sigma_2}^L &= e_{\mathbf{k}_1}^{(L)i} e_{\mathbf{k}_1}^{(L)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{q}}^{(\sigma_2)k} e_{\mathbf{q}}^{(\sigma_2)i*} = (e_{\mathbf{k}_1}^{(L)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(L)}) \\ &= A_{L\sigma_1} B_{\sigma_1 \sigma_2} C_{\sigma_2 L} \end{aligned} \quad (65)$$

where $\sigma_1, \sigma_2 = L, \pm$. For simplicity, we define:

$$A_{LL} = \frac{q \sin \theta \sin \theta_1 \cos \phi + (q \cos \theta + k_s) \cos \theta_1}{K}, \quad (66)$$

$$A_{L+} = \frac{\sin \theta_1 \cos \phi (q \cos \theta + k_s) - iK \sin \theta_1 \sin \phi - q \sin \theta \cos \theta_1}{\sqrt{2}K}, \quad (67)$$

$$A_{L-} = \frac{\sin \theta_1 \cos \phi (q \cos \theta + k_s) + iK \sin \theta_1 \sin \phi - q \sin \theta \cos \theta_1}{\sqrt{2}K}, \quad (68)$$

$$C_{LL} = \sin \theta \sin \theta_1 \cos \phi + \cos \theta \cos \theta_1, \quad (69)$$

$$C_{+L} = \frac{\sin \theta_1 (\cos \theta \cos \phi + i \sin \phi) - \sin \theta \cos \theta_1}{\sqrt{2}}, \quad (70)$$

$$C_{-L} = \frac{\sin \theta_1 (\cos \theta \cos \phi - i \sin \phi) - \sin \theta \cos \theta_1}{\sqrt{2}}, \quad (71)$$

and

$$B_{LL} = \frac{q + k_s \cos \theta}{K}, \quad B_{L+} = B_{L-} = -\frac{k_s \sin \theta}{\sqrt{2}K}, \quad (72)$$

$$B_{+L} = B_{-L} = \frac{k_s \sin \theta}{\sqrt{2}K}, \quad B_{++} = B_{--} = \frac{K + q + k_s \cos \theta}{2K}, \quad (73)$$

$$B_{+-} = B_{-+} = \frac{q + k_s \cos \theta - K}{2K}. \quad (74)$$

Thus, the nine contractions are:

$$\mathcal{E}_{LL}^L = A_{LL} B_{LL} C_{LL}, \quad \mathcal{E}_{L+}^L = A_{LL} B_{L+} C_{+L}, \quad \mathcal{E}_{L-}^L = A_{LL} B_{L-} C_{-L}, \quad (75)$$

$$\mathcal{E}_{+L}^L = A_{L+} B_{+L} C_{LL}, \quad \mathcal{E}_{++}^L = A_{L+} B_{++} C_{+L}, \quad \mathcal{E}_{+-}^L = A_{L+} B_{+-} C_{-L}, \quad (76)$$

$$\mathcal{E}_{-L}^L = A_{L-} B_{-L} C_{LL}, \quad \mathcal{E}_{-+}^L = A_{L-} B_{-+} C_{+L}, \quad \mathcal{E}_{--}^L = A_{L-} B_{--} C_{-L}. \quad (77)$$

We can then evaluate the loop integrals for the terms involving $D^{(L)}(k_1)$ and extract the nonlocal signal:

$$\mathcal{LOOP}_{\sigma_1 \sigma_2}^\lambda = \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D^{(\lambda)}(k_1, \tau_1, \tau_2)]_{ab} D^{(\sigma_1)}(K, \tau_2, \tau_3) D^{(\sigma_2)}(q, \tau_3, \tau_1) \cdot \mathcal{E}_{\sigma_1 \sigma_2}^L \quad (78)$$

$$= [D^{(\lambda)}(k_1, \tau_1, \tau_2)]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \times \frac{\Gamma[-id\tilde{\nu}]^4}{(4\pi)^2} \beta^{\sigma_1 L} \beta^{\sigma_2 L} \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2}\right)^{2id\tilde{\nu}} \left(\frac{K}{2}\right)^{2id\tilde{\nu}} \mathcal{E}_{\sigma_1 \sigma_2}^L \quad (79)$$

where $\beta^{\sigma L}$ is an additional factor for the longitudinal mode:

$$\beta^{\sigma L} = \begin{cases} \frac{(-1/2+id\tilde{\nu})^2}{\tilde{\nu}^2+1/4}, & \sigma = L, \\ 1, & \sigma = \pm. \end{cases} \quad (80)$$

Since all terms proportional to $\sin \phi$ or $\cos \phi$ vanish after the azimuthal integral, we only need the ϕ -averaged angular factors. For the azimuthal average, it is important to keep the quadratic terms:

$$\overline{\sin \phi} = \overline{\cos \phi} = \overline{\sin \phi \cos \phi} = 0, \quad \overline{\sin^2 \phi} = \overline{\cos^2 \phi} = \frac{1}{2}. \quad (81)$$

We denote $z = \cos \theta$, $s_1 = \sin \theta_1$, and $c_1 = \cos \theta_1$. The nonvanishing parts are:

$$\bar{\mathcal{E}}_{LL}^L = \frac{q + k_s z}{K^2} \left[\frac{1}{2} q (1 - z^2) s_1^2 + z (q z + k_s) c_1^2 \right], \quad (82)$$

$$\bar{\mathcal{E}}_{L+}^L = \bar{\mathcal{E}}_{L-}^L = -\frac{k_s (1 - z^2)}{2K^2} \left[\frac{1}{2} q z s_1^2 - (q z + k_s) c_1^2 \right], \quad (83)$$

$$\bar{\mathcal{E}}_{+L}^L = \bar{\mathcal{E}}_{-L}^L = \frac{k_s (1 - z^2)}{2K^2} \left[\frac{1}{2} (q z + k_s) s_1^2 - q z c_1^2 \right], \quad (84)$$

$$\bar{\mathcal{E}}_{++}^L = \bar{\mathcal{E}}_{--}^L = \frac{K + q + k_s z}{4K^2} \left[\frac{1}{2} (z (q z + k_s) + K) s_1^2 + q (1 - z^2) c_1^2 \right], \quad (85)$$

$$\bar{\mathcal{E}}_{+-}^L = \bar{\mathcal{E}}_{-+}^L = \frac{q + k_s z - K}{4K^2} \left[\frac{1}{2} (z (q z + k_s) - K) s_1^2 + q (1 - z^2) c_1^2 \right]. \quad (86)$$

For any monomial generated by these angular factors, we first define the dimensionful integral:

$$\begin{aligned} \tilde{\mathcal{J}}_r(m, n) &\equiv \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2} \right)^{2id\tilde{\nu}} \left(\frac{K}{2} \right)^{2id\tilde{\nu}} q^m K^{-n} z^r \\ &= \frac{4^{-2id\tilde{\nu}} k_s^{3+4id\tilde{\nu}+m-n}}{(4\pi)^{3/2}} \mathcal{J}_r(m, n), \end{aligned} \quad (87)$$

where the dimensionless loop seeds are:

$$\mathcal{J}_r(m, n) \equiv f_r \left(x - \frac{m}{2}, x + \frac{n}{2} \right), \quad x = -id\tilde{\nu}, \quad (88)$$

and $f_1 \equiv f_c$, $f_2 \equiv f_{cc}$. The $\mathcal{J}_3$ terms below are reduced to the same loop seeds by using $z = (K^2 - q^2 - k_s^2)/(2qk_s)$:

$$\begin{aligned} \mathcal{J}_3(m, n) &= \frac{1}{8} [\mathcal{J}_0(m-3, n-6) - 3\mathcal{J}_0(m-1, n-4) - 3\mathcal{J}_0(m-3, n-4) \\ &\quad + 3\mathcal{J}_0(m+1, n-2) + 6\mathcal{J}_0(m-1, n-2) + 3\mathcal{J}_0(m-3, n-2) \\ &\quad - \mathcal{J}_0(m+3, n) - 3\mathcal{J}_0(m+1, n) - 3\mathcal{J}_0(m-1, n) - \mathcal{J}_0(m-3, n)]. \end{aligned} \quad (89)$$

We denote the dimensionless coefficients of the loop integrals by $\mathcal{I}_{\sigma_1\sigma_2}^\lambda(-id\tilde{\nu})$ and define them through:

$$\mathcal{LOOP}_{\sigma_1\sigma_2}^\lambda = \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \frac{\Gamma^4(-id\tilde{\nu})}{(4\pi)^2} \tilde{\mathcal{I}}_{\sigma_1\sigma_2}^\lambda(-id\tilde{\nu}) \quad (90)$$

and explicitly:

$$\begin{aligned} \tilde{\mathcal{I}}_{\sigma_1\sigma_2}^\lambda &\equiv \beta^{\sigma_1 L} \beta^{\sigma_2 L} \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2} \right)^{2id\tilde{\nu}} \left(\frac{K}{2} \right)^{2id\tilde{\nu}} \bar{\mathcal{E}}_{\sigma_1\sigma_2}^\lambda \\ &= \frac{4^{-2id\tilde{\nu}} k_s^{3+4id\tilde{\nu}}}{(4\pi)^{3/2}} \mathcal{I}_{\sigma_1\sigma_2}^\lambda, \quad d = \pm. \end{aligned} \quad (91)$$

Then the dimensionless coefficients of the nine loop integrals are:

$$\mathcal{I}_{LL}^L = \frac{(-1/2 + id\tilde{\nu})^4}{(\tilde{\nu}^2 + 1/4)^2} \cdot \left\{ \frac{s_1^2}{2} [\mathcal{J}_0(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_2(2, 2) - \mathcal{J}_3(1, 2)] \right. \\ \left. + c_1^2 [\mathcal{J}_2(2, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_1(1, 2) + \mathcal{J}_2(0, 2)] \right\}, \quad (92)$$

$$\mathcal{I}_{L+}^L = \mathcal{I}_{L-}^L = \frac{(-1/2 + id\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \cdot \left\{ -\frac{s_1^2}{4} [\mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \right. \\ \left. + \frac{c_1^2}{2} [\mathcal{J}_1(1, 2) + \mathcal{J}_0(0, 2) - \mathcal{J}_3(1, 2) - \mathcal{J}_2(0, 2)] \right\}, \quad (93)$$

$$\mathcal{I}_{+L}^L = \mathcal{I}_{-L}^L = \frac{(-1/2 + id\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \cdot \left\{ \frac{s_1^2}{4} [\mathcal{J}_1(1, 2) + \mathcal{J}_0(0, 2) - \mathcal{J}_3(1, 2) - \mathcal{J}_2(0, 2)] \right. \\ \left. - \frac{c_1^2}{2} [\mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \right\}, \quad (94)$$

$$\mathcal{I}_{++}^L = \mathcal{I}_{--}^L = \frac{s_1^2}{8} [\mathcal{J}_2(1, 1) + \mathcal{J}_2(2, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_1(0, 1) + \mathcal{J}_1(1, 2) + \mathcal{J}_2(0, 2) \\ + \mathcal{J}_0(0, 0) + \mathcal{J}_0(1, 1) + \mathcal{J}_1(0, 1)] \\ + \frac{c_1^2}{4} [\mathcal{J}_0(1, 1) - \mathcal{J}_2(1, 1) + \mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)], \quad (95)$$

$$\mathcal{I}_{+-}^L = \mathcal{I}_{-+}^L = \frac{s_1^2}{8} [\mathcal{J}_2(2, 2) + \mathcal{J}_3(1, 2) - \mathcal{J}_2(1, 1) + \mathcal{J}_1(1, 2) + \mathcal{J}_2(0, 2) - \mathcal{J}_1(0, 1) \\ - \mathcal{J}_0(1, 1) - \mathcal{J}_1(0, 1) + \mathcal{J}_0(0, 0)] \\ + \frac{c_1^2}{4} [\mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2) - \mathcal{J}_0(1, 1) + \mathcal{J}_2(1, 1)]. \quad (96)$$

Thus, this part of the triangle loop integral can be written as:

$$\mathcal{LOOP}^L = \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{LOOP}_{\sigma_1\sigma_2}^L \\ = \left[D^{(L)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \\ \times \frac{4^{-2id\tilde{\nu}} k_s^{3+4id\tilde{\nu}} \Gamma^4(-id\tilde{\nu})}{(4\pi)^{7/2}} \mathcal{I}^L, \quad (97)$$

where $\mathcal{I}^L$ is defined as the sum of the nine contractions:

$$\mathcal{I}^L \equiv \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1\sigma_2}^L = (\mathcal{I}_{LL}^L + 2\mathcal{I}_{L+}^L + 2\mathcal{I}_{+L}^L + 2\mathcal{I}_{++}^L + 2\mathcal{I}_{+-}^L) \quad (98)$$

Equivalently,

$$\mathcal{LOOP}^L = \left[D^{(L)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \mathcal{B}_d^L, \quad (99)$$

where

$$\mathcal{B}_d^L = \frac{4^{-2id\tilde{\nu}} k_s^{3+4id\tilde{\nu}} \Gamma^4(-id\tilde{\nu})}{(4\pi)^{7/2}} \mathcal{I}^{(L)}. \quad (100)$$

We now repeat the same calculation for the $\lambda = \pm$ terms. The vector contractions are:

$$\begin{aligned}\mathcal{E}_{\sigma_1\sigma_2}^{(\sigma)} &= e_{\mathbf{k}_1}^{(\sigma)i} e_{\mathbf{k}_1}^{(\sigma)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{q}}^{(\sigma_2)k} e_{\mathbf{q}}^{(\sigma_2)i*} \\ &= (e_{\mathbf{k}_1}^{(\sigma)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\sigma)}), \quad \sigma = \pm.\end{aligned}\quad (101)$$

For the first and third factors, we define:

$$\begin{aligned}A_{+L} &= \frac{q \sin \theta \cos \phi \cos \theta_1 - iq \sin \theta \sin \phi - (q \cos \theta + k_s) \sin \theta_1}{\sqrt{2}K}, \\ A_{++} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi - iK \sin \phi \cos \theta_1 - i \sin \phi (q \cos \theta + k_s) + K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\ A_{+-} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi + iK \sin \phi \cos \theta_1 - i \sin \phi (q \cos \theta + k_s) - K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\ C_{L+} &= \frac{\sin \theta \cos \phi \cos \theta_1 + i \sin \theta \sin \phi - \cos \theta \sin \theta_1}{\sqrt{2}}, \\ C_{++} &= \frac{\cos \theta \cos \theta_1 \cos \phi + i \sin \phi \cos \theta_1 + i \cos \theta \sin \phi + \cos \phi + \sin \theta \sin \theta_1}{2}, \\ C_{-+} &= \frac{\cos \theta \cos \theta_1 \cos \phi - i \sin \phi \cos \theta_1 + i \cos \theta \sin \phi - \cos \phi + \sin \theta \sin \theta_1}{2}.\end{aligned}\quad (103)$$

The $D^{(-)}(k_1)$ factors are obtained by complex conjugating the external polarization:

$$\begin{aligned}A_{-L} &= \frac{q \sin \theta \cos \phi \cos \theta_1 + iq \sin \theta \sin \phi - (q \cos \theta + k_s) \sin \theta_1}{\sqrt{2}K}, \\ A_{-+} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi - iK \sin \phi \cos \theta_1 + i \sin \phi (q \cos \theta + k_s) - K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\ A_{--} &= \frac{(q \cos \theta + k_s) \cos \theta_1 \cos \phi + iK \sin \phi \cos \theta_1 + i \sin \phi (q \cos \theta + k_s) + K \cos \phi + q \sin \theta \sin \theta_1}{2K}, \\ C_{L-} &= \frac{\sin \theta \cos \phi \cos \theta_1 - i \sin \theta \sin \phi - \cos \theta \sin \theta_1}{\sqrt{2}}, \\ C_{+-} &= \frac{\cos \theta \cos \theta_1 \cos \phi + i \sin \phi \cos \theta_1 - i \cos \theta \sin \phi - \cos \phi + \sin \theta \sin \theta_1}{2}, \\ C_{--} &= \frac{\cos \theta \cos \theta_1 \cos \phi - i \sin \phi \cos \theta_1 - i \cos \theta \sin \phi + \cos \phi + \sin \theta \sin \theta_1}{2}.\end{aligned}\quad (104)$$

The middle factor $B_{\sigma_1\sigma_2}$ is the same as before. Therefore, the nine contractions are:

$$\mathcal{E}_{LL}^\sigma = A_{\sigma L} B_{LL} C_{L\sigma}, \quad \mathcal{E}_{L+}^\sigma = A_{\sigma L} B_{L+} C_{+\sigma}, \quad \mathcal{E}_{L-}^\sigma = A_{\sigma L} B_{L-} C_{-\sigma}, \quad (105)$$

$$\mathcal{E}_{+L}^\sigma = A_{\sigma+} B_{+L} C_{L\sigma}, \quad \mathcal{E}_{++}^\sigma = A_{\sigma+} B_{++} C_{+\sigma}, \quad \mathcal{E}_{+-}^\sigma = A_{\sigma+} B_{+-} C_{-\sigma}, \quad (106)$$

$$\mathcal{E}_{-L}^\sigma = A_{\sigma-} B_{-L} C_{L\sigma}, \quad \mathcal{E}_{-+}^\sigma = A_{\sigma-} B_{-+} C_{+\sigma}, \quad \mathcal{E}_{--}^\sigma = A_{\sigma-} B_{--} C_{-\sigma}. \quad (107)$$

After taking the azimuthal average, we obtain the nonvanishing parts of the nine contractions:

$$\begin{aligned}
\bar{\mathcal{E}}_{LL}^+ &= \frac{q + k_s z}{2K^2} \left[\frac{q(1 + c_1^2)}{2} (1 - z^2) + (qz + k_s) z s_1^2 \right] \\
\bar{\mathcal{E}}_{L+}^+ &= \left(-\frac{k_s(1 - z^2)}{4K^2} \right) \left[qc_1 + qz \frac{1 + c_1^2}{2} - (qz + k_s) s_1^2 \right] \\
\bar{\mathcal{E}}_{L-}^+ &= \left(-\frac{k_s(1 - z^2)}{4K^2} \right) \left[-qc_1 + qz \frac{1 + c_1^2}{2} - (qz + k_s) s_1^2 \right] \\
\bar{\mathcal{E}}_{+L}^+ &= \left(\frac{k_s(1 - z^2)}{4K^2} \right) \left[Kc_1 + (qz + k_s) \frac{1 + c_1^2}{2} - qz s_1^2 \right] \\
\bar{\mathcal{E}}_{-L}^+ &= \left(\frac{k_s(1 - z^2)}{4K^2} \right) \left[-Kc_1 + (qz + k_s) \frac{1 + c_1^2}{2} - qz s_1^2 \right] \\
\bar{\mathcal{E}}_{++}^+ &= \left(\frac{K + q + k_s z}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z + 2c_1 + z}{2} + K \frac{c_1^2 + 2zc_1 + 1}{2} + qs_1^2(1 - z^2) \right] \\
\bar{\mathcal{E}}_{--}^+ &= \left(\frac{K + q + k_s z}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z - 2c_1 + z}{2} + K \frac{c_1^2 - 2zc_1 + 1}{2} + qs_1^2(1 - z^2) \right] \\
\bar{\mathcal{E}}_{+-}^+ &= \left(\frac{q + k_s z - K}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z - 2c_1 + z}{2} - K \frac{c_1^2 - 2zc_1 + 1}{2} + qs_1^2(1 - z^2) \right] \\
\bar{\mathcal{E}}_{-+}^+ &= \left(\frac{q + k_s z - K}{8K^2} \right) \left[(qz + k_s) \frac{c_1^2 z + 2c_1 + z}{2} - K \frac{c_1^2 + 2zc_1 + 1}{2} + qs_1^2(1 - z^2) \right] \tag{108}
\end{aligned}$$

Applying the same loop-seed map as above, we obtain the dimensionless coefficients:

$$\begin{aligned}
\mathcal{I}_{LL}^+ &= \frac{(-1/2 + i\tilde{d}\tilde{\nu})^4}{(\tilde{\nu}^2 + 1/4)^2} \cdot \left\{ \frac{1 + c_1^2}{4} [\mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \right. \\
&\quad \left. + \frac{s_1^2}{2} [\mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_2(0, 2)] \right\} \\
\mathcal{I}_{L+}^+ + \mathcal{I}_{L-}^+ + \mathcal{I}_{+L}^+ + \mathcal{I}_{-L}^+ &= \frac{(-1/2 + i\tilde{d}\tilde{\nu})^2}{(\tilde{\nu}^2 + 1/4)} \frac{2 + s_1^2}{4} [\mathcal{J}_0(0, 2) - \mathcal{J}_2(0, 2)] \\
\mathcal{I}_{++}^+ + \mathcal{I}_{--}^+ + \mathcal{I}_{+-}^+ + \mathcal{I}_{-+}^+ &= \frac{1 + c_1^2}{4} [\mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) + \mathcal{J}_3(1, 2) + \mathcal{J}_2(0, 2) + \mathcal{J}_0(0, 0)] \\
&\quad + \frac{s_1^2}{2} [\mathcal{J}_0(2, 2) - \mathcal{J}_2(2, 2) + \mathcal{J}_1(1, 2) - \mathcal{J}_3(1, 2)] \tag{109}
\end{aligned}$$

Thus, the sum of these nine contractions is:

$$\mathcal{I}^+ \equiv \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1 \sigma_2}^+ \tag{110}$$

For the $D^{(-)}(k_1)$ factors, the polarization contractions satisfy:

$$\bar{\mathcal{E}}_{LL}^- = \bar{\mathcal{E}}_{LL}^+, \quad \bar{\mathcal{E}}_{L+}^- = \bar{\mathcal{E}}_{L-}^+, \quad \bar{\mathcal{E}}_{L-}^- = \bar{\mathcal{E}}_{L+}^+, \tag{111}$$

$$\bar{\mathcal{E}}_{+L}^- = \bar{\mathcal{E}}_{-L}^+, \quad \bar{\mathcal{E}}_{-L}^- = \bar{\mathcal{E}}_{+L}^+, \quad \bar{\mathcal{E}}_{++}^- = \bar{\mathcal{E}}_{--}^+, \tag{112}$$

$$\bar{\mathcal{E}}_{--}^- = \bar{\mathcal{E}}_{++}^+, \quad \bar{\mathcal{E}}_{+-}^- = \bar{\mathcal{E}}_{-+}^+, \quad \bar{\mathcal{E}}_{-+}^- = \bar{\mathcal{E}}_{+-}^+. \tag{113}$$

Thus, the sum of these nine contractions is:

$$\mathcal{I}^- \equiv \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1 \sigma_2}^- = \mathcal{I}^+ \quad (114)$$

Thus, the transverse part of the triangle loop integral is:

$$\mathcal{LOOP}^\sigma = \left[D^{(\sigma)}(k_1, \tau_1, \tau_2) \right]_{ab} \cdot \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \mathcal{B}_d^\sigma, \quad (115)$$

with the bubble subgraph contribution:

$$\mathcal{B}_d^\sigma = \frac{\Gamma^4(-id\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2id\tilde{\nu}} k_s^3 4^{id\tilde{\nu}} \mathcal{I}^\sigma, \quad \sigma = \pm. \quad (116)$$

The total loop integral therefore becomes:

$$\mathcal{LOOP} = \sum_{\lambda=L, \pm} \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} \sum_{d=\pm} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} (-\tau_3)^{1+2id\tilde{\nu}} \mathcal{B}_d^\lambda \quad (117)$$

We next evaluate the contributions from the left and right subgraphs:

$$\begin{aligned} \mathcal{T}_d^{(L), \lambda} &= (g^2 h_0)^2 \sum_{a, b=\pm} (-ab) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} G_a(k_1; \tau_1) G_b(k_2; \tau_2) \\ &\quad \times \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} (\tau_1 \tau_2)^{1/2+id\tilde{\nu}} \end{aligned} \quad (118)$$

$$\begin{aligned} &= (g^2 h_0)^2 \frac{1}{4k_1^3 k_2^3} \sum_{a, b=\pm} (-ab) \int d\tau_1 d\tau_2 (1 - iak_1\tau_1)(1 - ibk_2\tau_2) \\ &\quad \times \left[D^{(\lambda)}(k_1, \tau_1, \tau_2) \right]_{ab} (-\tau_1)^{-3/2+id\tilde{\nu}} (-\tau_2)^{-3/2+id\tilde{\nu}} e^{iak_1\tau_1 + ibk_2\tau_2} \end{aligned} \quad (119)$$

For $\lambda = L$, this integral is given by the tree-level seed of Ref. [1]:

$$\mathcal{T}_d^{(L), L} = (g^2 h_0)^2 \frac{k_{12}^{-2id\tilde{\nu}}}{4k_1^3 k_2^3} \cdot \mathcal{P}_d^{(2), L}(\tilde{\nu}) \quad (120)$$

where the coefficient is:

$$\begin{aligned} \mathcal{P}_d^{(2), L}(\tilde{\nu}) &= 2^{2id\tilde{\nu}} \cdot \sum_{a, b=\pm} \left[+\mathcal{K}_{ab}^{-\frac{3}{2}+id\tilde{\nu}, -\frac{3}{2}+id\tilde{\nu}} + ia\mathcal{K}_{ab}^{-\frac{1}{2}+id\tilde{\nu}, -\frac{3}{2}+id\tilde{\nu}} \right. \\ &\quad \left. + ib\mathcal{K}_{ab}^{-\frac{3}{2}+id\tilde{\nu}, -\frac{1}{2}+id\tilde{\nu}} - ab\mathcal{K}_{ab}^{-\frac{1}{2}+id\tilde{\nu}, -\frac{1}{2}+id\tilde{\nu}} \right] \end{aligned} \quad (121)$$

Here, $\mathcal{K}_{ab}^{p_1 p_2} = \mathcal{I}_{ab}^{(L)p_1 p_2}(1, 1)$ are the vector seeds in the double-folded limit [1]. The opposite-sign seeds are:

$$\begin{aligned} \mathcal{K}_{\pm\mp}^{p_1 p_2} &= \frac{e^{\mp i\pi(p_1 - p_2)/2}}{4\pi(\tilde{\nu}^2 + \frac{1}{4})} \left[\mathbf{H}_{\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2}\right)^{i\tilde{\nu}} + \mathbf{H}_{-\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2}\right)^{-i\tilde{\nu}} \right] \\ &\quad \times \left[\mathbf{H}_{\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2}\right)^{i\tilde{\nu}} + \mathbf{H}_{-\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2}\right)^{-i\tilde{\nu}} \right] \end{aligned} \quad (122)$$

where we define

$$\mathbf{H}_{\tilde{\nu}}^p(1) \equiv \left\{ \left(\frac{1}{2} - i\tilde{\nu} \right) \Gamma \left[-i\tilde{\nu}, p + \frac{3}{2} + i\tilde{\nu} \right] {}_2F_1 \left[\begin{matrix} \frac{p}{2} + \frac{3}{4} + \frac{i\tilde{\nu}}{2}, & \frac{p}{2} + \frac{5}{4} + \frac{i\tilde{\nu}}{2} \\ 1 + i\tilde{\nu} \end{matrix} \middle| 1 \right] \right. \\ \left. + \frac{1}{2} \Gamma \left[-1 - i\tilde{\nu}, p + \frac{7}{2} + i\tilde{\nu} \right] {}_2F_1 \left[\begin{matrix} \frac{p}{2} + \frac{7}{4} + \frac{i\tilde{\nu}}{2}, & \frac{p}{2} + \frac{9}{4} + \frac{i\tilde{\nu}}{2} \\ 2 + i\tilde{\nu} \end{matrix} \middle| 1 \right] \right\} \quad (123)$$

The same-sign seeds are:

$$\mathcal{K}_{\pm\pm}^{p_1 p_2} = \frac{-ie^{\mp i\pi(p_1+p_2)/2}}{4\pi(\tilde{\nu}^2 + \frac{1}{4})} \left[e^{\pi\tilde{\nu}} \mathbf{H}_{\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2} \right)^{i\tilde{\nu}} + e^{-\pi\tilde{\nu}} \mathbf{H}_{-\tilde{\nu}}^{p_1}(1) \left(\frac{1}{2} \right)^{-i\tilde{\nu}} \right] \left[\mathbf{H}_{\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2} \right)^{i\tilde{\nu}} + \mathbf{H}_{-\tilde{\nu}}^{p_2}(1) \left(\frac{1}{2} \right)^{-i\tilde{\nu}} \right] \\ + \frac{e^{\mp i\pi(p_1+p_2)/2}}{2(\tilde{\nu}^2 + \frac{1}{4})} \sum_{n_1, n_2=0}^{\infty} \frac{(-1)^{n_{12}}}{n_1! n_2!} \left(\frac{1}{2} \right)^{2n_{12}} \frac{i}{\tilde{\nu}} \left(2n_1 - \frac{1}{2} + i\tilde{\nu} \right) \left(2n_2 - \frac{1}{2} - i\tilde{\nu} \right) (-i\tilde{\nu})_{-n_1} (i\tilde{\nu})_{-n_2} \\ \times \mathcal{F} \left[\begin{matrix} 2n_2 + p_2 + \frac{3}{2} - i\tilde{\nu}, & 2n_{12} + p_1 + p_2 + 3 \\ 2n_2 + p_2 + \frac{5}{2} - i\tilde{\nu} \end{matrix} \middle| -1 \right] + (\tilde{\nu} \rightarrow -\tilde{\nu}) \quad (124)$$

However, the double-folded limit is difficult to evaluate in this form. We therefore use another method, namely, applying a differential operator to the scalar seed [2]. The coefficient then decomposes into two parts, the first of which contains the delta-function contribution:

$$\text{Part 1 of } \mathcal{P}_d^{(2),L}(\tilde{\nu}) = \mathcal{P}_d^{(1)}(\tilde{\nu}) \cdot \frac{1}{\tilde{\nu}^2 + 1/4} \quad (125)$$

The second part arises from the scalar seeds:

$$\text{Part 2 of } \mathcal{P}_d^{(2),L}(\tilde{\nu}) = \frac{2^{2id\tilde{\nu}}}{\tilde{\nu}^2 + 1/4} \sum_{a,b=\pm} \left\{ \left(\frac{1}{2} + id\tilde{\nu} \right)^2 \left[\mathcal{M}_{ab}^{p_0, p_0} + (ia) \mathcal{M}_{ab}^{p_0+1, p_0} + (ib) \mathcal{M}_{ab}^{p_0, p_0+1} - (ab) \mathcal{M}_{ab}^{p_0+1, p_0+1} \right] \right. \\ \left. + \left(\frac{1}{2} + id\tilde{\nu} \right) \left[\mathcal{M}_{ab}^{p_0+2, p_0} + (ib) \mathcal{M}_{ab}^{p_0+2, p_0+1} + \mathcal{M}_{ab}^{p_0, p_0+2} + (ia) \mathcal{M}_{ab}^{p_0+1, p_0+2} \right] \right. \\ \left. + \mathcal{M}_{ab}^{p_0+2, p_0+2} \right\} \quad (126)$$

where $p_0 = -\frac{5}{2} + id\tilde{\nu}$, and $\mathcal{M}_{ab}^{p_1 p_2} = \mathcal{I}_{ab}^{p_1 p_2}(1, 1)$ are the scalar seeds defined in [2]. Since p_0 lies on the boundary of the seed convergence domain, the expressions are understood by analytic continuation from $p_0 = -\frac{5}{2} + \epsilon + id\tilde{\nu}$, taking the finite part as $\epsilon \rightarrow 0^+$. The opposite-sign seeds are:

$$\mathcal{M}_{\pm\mp}^{p_1 p_2} = \frac{e^{\mp i\tilde{p}_{12}\pi/2}}{2^{5+p_{12}}} \Gamma \left[\begin{matrix} \frac{5}{2} + p_1 - i\tilde{\nu}, & \frac{5}{2} + p_1 + i\tilde{\nu}, & \frac{5}{2} + p_2 - i\tilde{\nu}, & \frac{5}{2} + p_2 + i\tilde{\nu} \\ 3 + p_1, & 3 + p_2 \end{matrix} \right] \quad (127)$$

The same-sign seeds are:

$$\mathcal{M}_{\pm\pm}^{p_1 p_2} = \frac{\pm i e^{\mp i p_{12}\pi/2} e^{-\pi\tilde{\nu}}}{2^{5+p_{12}}} \Gamma \left[\begin{matrix} \frac{5}{2} + p_1 - i\tilde{\nu}, & \frac{5}{2} + p_1 + i\tilde{\nu}, & \frac{5}{2} + p_2 - i\tilde{\nu}, & \frac{5}{2} + p_2 + i\tilde{\nu} \\ 3 + p_1, & 3 + p_2 \end{matrix} \right] \\ - \frac{e^{\mp i p_{12}\pi/2}}{2^{5+p_{12}}} \Gamma \left[5 + p_{12}, \frac{5}{2} + p_1 \pm i\tilde{\nu}, \frac{5}{2} + p_2 \pm i\tilde{\nu} \right] \\ \cdot {}_3\tilde{F}_2 \left[\begin{matrix} 5 + p_{12}, & \frac{1}{2} \pm i\tilde{\nu}, & 1 \\ \frac{7}{2} + p_1 \pm i\tilde{\nu}, & \frac{7}{2} + p_2 \pm i\tilde{\nu} \end{matrix} \middle| 1 \right]. \quad (128)$$

For $\lambda = \pm$, this integral can also be expressed in terms of the tree-level seed of Ref. [2]:

$$\begin{aligned}\mathcal{T}_d^{(L),\pm} &= (g^2 h_0)^2 \frac{1}{4H^2 k_1^3 k_2^3} \sum_{a,b=\pm} (-ab) \int d\tau_1 d\tau_2 (1 - iak_1\tau_1)(1 - ibk_2\tau_2) \\ &\times \left[D^{(\text{scalar})}(k_1, \tau_1, \tau_2) \right]_{ab} (-\tau_1)^{-5/2+id\tilde{\nu}} (-\tau_2)^{-5/2+id\tilde{\nu}} e^{iak_1\tau_1+ibk_2\tau_2}\end{aligned}\quad (129)$$

$$= (g^2 h_0)^2 \frac{k_{12}^{-2id\tilde{\nu}}}{4k_1^3 k_2^3} \cdot \mathcal{P}_d^{(2),\pm}(\tilde{\nu}) \quad (130)$$

where the coefficient is:

$$\mathcal{P}_d^{(2),\pm}(\tilde{\nu}) = 2^{2id\tilde{\nu}} \cdot \sum_{a,b=\pm} \left[+\mathcal{M}_{ab}^{p_0,p_0} + ia\mathcal{M}_{ab}^{p_0+1,p_0} + ib\mathcal{M}_{ab}^{p_0,p_0+1} - ab\mathcal{M}_{ab}^{p_0+1,p_0+1} \right] \quad (131)$$

Similarly, for the right subgraph:

$$\begin{aligned}\mathcal{T}_d^{(R)} &= (g^2) \sum_c ic \int \frac{d\tau_3}{|H\tau_3|^2} G_c(k_3; \tau_3) G_c(k_4; \tau_3) (-\tau_3)^{1+2id\tilde{\nu}} \\ &= (g^2) \frac{H^2}{4k_3^3 k_4^3} \sum_{c=\pm} ic \int_{-\infty}^0 d\tau_3 (-\tau_3)^{-1+2id\tilde{\nu}} (1 - ick_{34}\tau_3 - k_3 k_4 \tau_3^2) e^{ick_{34}\tau_3} \\ &= (g^2) \frac{H^2}{4k_3^3 k_4^3} \times k_{34}^{-2id\tilde{\nu}} \cdot \mathcal{P}_d^{(1)}(\tilde{\nu})\end{aligned}\quad (132)$$

where the coefficient is:

$$\mathcal{P}_d^{(1)}(\tilde{\nu}) = \frac{2 + id\tilde{\nu}}{2id\tilde{\nu}} i \sinh(\pi d\tilde{\nu}) \Gamma(2 + 2id\tilde{\nu}) \quad (133)$$

Thus, the contribution of one triangle diagram is:

$$\langle \delta h^4 \rangle^{\text{Tri1}} = \sum_{d=\pm} \sum_{\lambda=L,\pm} \mathcal{T}_d^{(L),\lambda} \cdot \mathcal{B}_d^\lambda \cdot \mathcal{T}_d^{(R)} \quad (134)$$

$$= (g^2)(g^2 h_0)^2 \frac{H^2 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{\text{Tri}}(\tilde{\nu}, \theta_1) + \text{c.c.} \quad (135)$$

where the magnitude is:

$$C^{\text{Tri}}(\tilde{\nu}, \theta_1) = \sum_{\lambda=L,\pm} [\mathcal{P}_+^{(1)}(\tilde{\nu}) \mathcal{P}_+^{(2),\lambda}(\tilde{\nu})] \cdot \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot \mathcal{I}^\lambda(\tilde{\nu}, \theta_1)|_{d=+} \quad (136)$$

For the second triangle diagram:

$$\langle \delta h^4 \rangle^{\text{Tri2}} = \langle \delta h^4 \rangle^{\text{Tri1}}|_{k_1 \longleftrightarrow k_3, k_2 \longleftrightarrow k_4, \theta_1 \longleftrightarrow \theta_3}, \quad (137)$$

Thus, the total contribution of the triangle diagrams is:

$$\begin{aligned}\langle \delta h^4 \rangle^{\text{Tri}} &= \langle \delta h^4 \rangle^{\text{Tri1}} + \langle \delta h^4 \rangle^{\text{Tri2}} \\ &= (g^2)(g^2 h_0)^2 \frac{H^2 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times (C^{\text{Tri}}(\tilde{\nu}, \theta_1) + C^{\text{Tri}}(\tilde{\nu}, \theta_3)) + \text{c.c.}\end{aligned}\quad (138)$$

A numerical result is shown below:

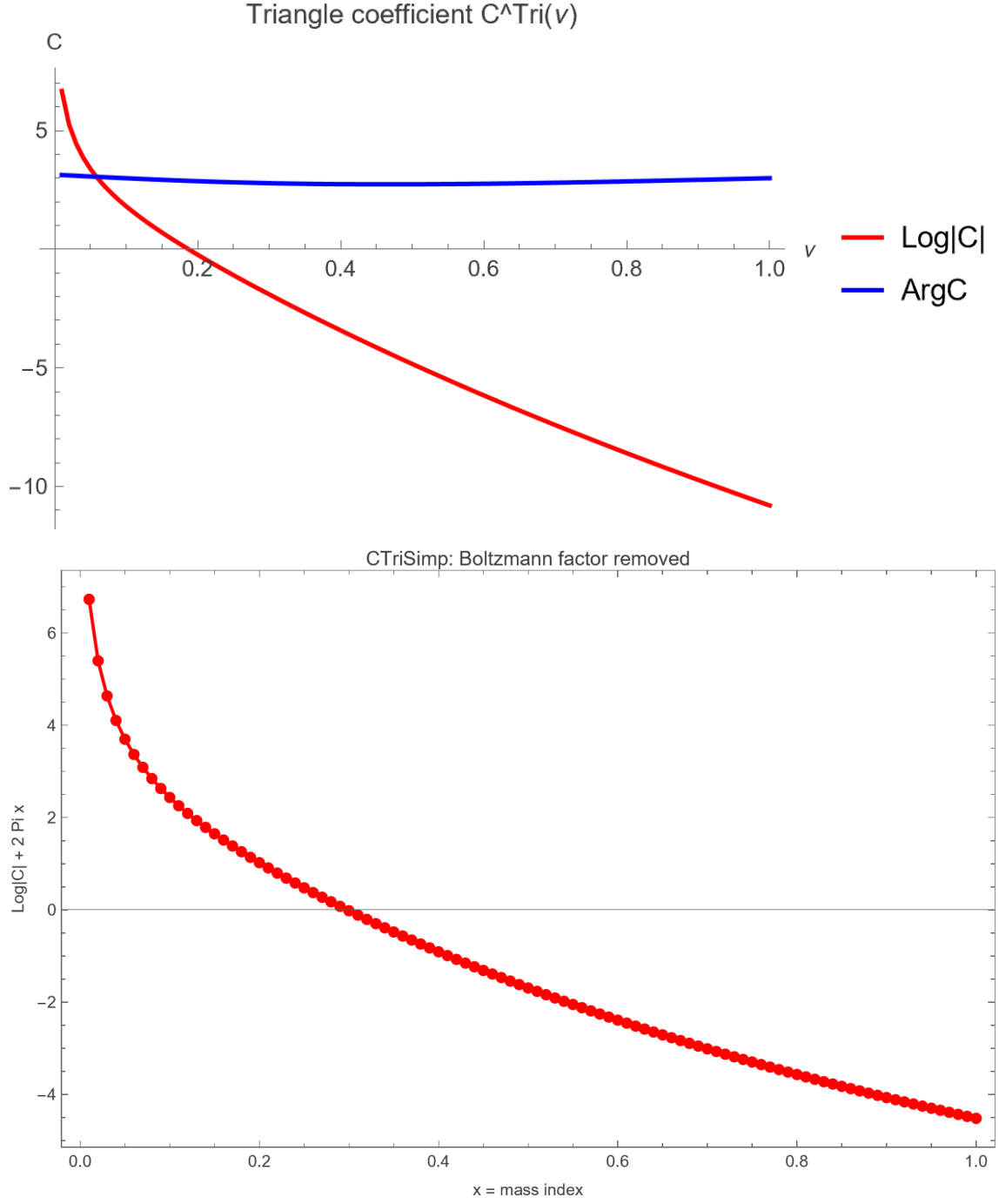


Figure 2: $C_{\text{Tri}}(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |C_{\text{Tri}}(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom), with the leading $e^{-2\pi\tilde{\nu}}$ factor divided out.

4 Box Diagram

For the box diagram, we have:

$$\begin{aligned}
\langle \delta h^4 \rangle^{\text{Box}} &= (ig^2 h_0)^4 \sum_{a,b,c,d=\pm} (abcd) \int \frac{d\tau_1}{|H\tau_1|^2} \frac{d\tau_2}{|H\tau_2|^2} \frac{d\tau_3}{|H\tau_3|^2} \frac{d\tau_4}{|H\tau_4|^2} \\
&\times G_a(k_1; \tau_1) G_b(k_2; \tau_2) G_c(k_3; \tau_3) G_d(k_4; \tau_4) \\
&\times \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} \\
&\times [D_{kl}(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4)]_{cd} [D_{li}(\mathbf{q}, \tau_4, \tau_1)]_{da}
\end{aligned} \tag{139}$$

The relevant loop integral is:

$$\begin{aligned}
\mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{q} + \mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)]_{bc} \\
&\times [D_{kl}(\mathbf{q} + \mathbf{k}_3, \tau_3, \tau_4)]_{cd} [D_{li}(\mathbf{q}, \tau_4, \tau_1)]_{da}
\end{aligned} \tag{140}$$

The cutting rule gives:

$$\begin{aligned}
\mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} [D_{ij}(\mathbf{k}_1, \tau_1, \tau_2)]_{ab} [D_{jk}(\mathbf{q} + \mathbf{k}_s, \tau_2, \tau_3)] \\
&\times [D_{kl}(\mathbf{k}_3, \tau_3, \tau_4)]_{cd} [D_{li}(\mathbf{q}, \tau_4, \tau_1)]
\end{aligned} \tag{141}$$

Expanding the four propagators in the helicity basis, the loop integral becomes:

$$\begin{aligned}
\mathcal{LOOP} &= \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left[D^{(L)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(L)i} e_{\mathbf{k}_1}^{(L)j*} + D^{(\pm)}(k_1, \tau_1, \tau_2) e_{\mathbf{k}_1}^{(\pm)i} e_{\mathbf{k}_1}^{(\pm)j*} \right]_{ab} \\
&\times \left[D^{(L)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(L)j} e_{\mathbf{K}}^{(L)k*} + D^{(\pm)}(K, \tau_2, \tau_3) e_{\mathbf{K}}^{(\pm)j} e_{\mathbf{K}}^{(\pm)k*} \right] \\
&\times \left[D^{(L)}(k_3, \tau_3, \tau_4) e_{\mathbf{k}_3}^{(L)k} e_{\mathbf{k}_3}^{(L)l*} + D^{(\pm)}(k_3, \tau_3, \tau_4) e_{\mathbf{k}_3}^{(\pm)k} e_{\mathbf{k}_3}^{(\pm)l*} \right]_{cd} \\
&\times \left[D^{(L)}(q, \tau_4, \tau_1) e_{\mathbf{q}}^{(L)l} e_{\mathbf{q}}^{(L)i*} + D^{(\pm)}(q, \tau_4, \tau_1) e_{\mathbf{q}}^{(\pm)l} e_{\mathbf{q}}^{(\pm)i*} \right],
\end{aligned} \tag{142}$$

where $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, and each repeated $\pm$ label is summed over the two transverse helicities.

To organize the nine external-helicity combinations, we choose the same convention for $\mathbf{k}_1$ as in the triangle diagram and set

$$\mathbf{k}_1 = k_1(\sin \theta_1, 0, \cos \theta_1), \tag{143}$$

$$\mathbf{k}_3 = k_3(\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3). \tag{144}$$

Let $\lambda_1, \lambda_2 = L, \pm$ label the polarizations of the hard propagators with momenta $\mathbf{k}_1$ and $\mathbf{k}_3$, respectively, and let $\sigma_1, \sigma_2 = L, \pm$ label the two soft propagators. Keeping the latter two helicities explicit, as in the triangle calculation, we define

$$\begin{aligned}
\mathcal{E}_{\sigma_1 \sigma_2}^{\lambda_1 \lambda_2} &\equiv e_{\mathbf{k}_1}^{(\lambda_1)i} e_{\mathbf{k}_1}^{(\lambda_1)j*} e_{\mathbf{K}}^{(\sigma_1)j} e_{\mathbf{K}}^{(\sigma_1)k*} e_{\mathbf{k}_3}^{(\lambda_2)k} e_{\mathbf{k}_3}^{(\lambda_2)l*} e_{\mathbf{q}}^{(\sigma_2)l} e_{\mathbf{q}}^{(\sigma_2)i*} \\
&= (e_{\mathbf{k}_1}^{(\lambda_1)*} \cdot e_{\mathbf{K}}^{(\sigma_1)}) (e_{\mathbf{K}}^{(\sigma_1)*} \cdot e_{\mathbf{k}_3}^{(\lambda_2)}) (e_{\mathbf{k}_3}^{(\lambda_2)*} \cdot e_{\mathbf{q}}^{(\sigma_2)}) (e_{\mathbf{q}}^{(\sigma_2)*} \cdot e_{\mathbf{k}_1}^{(\lambda_1)}).
\end{aligned} \tag{145}$$

The dimensionful loop integral is defined by

$$\begin{aligned}\tilde{\mathcal{I}}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2} &\equiv \beta^{\sigma_1 L} \beta^{\sigma_2 L} \int \frac{d\mathbf{q}^3}{(2\pi)^3} \left(\frac{q}{2}\right)^{2if\tilde{\nu}} \left(\frac{K}{2}\right)^{2if\tilde{\nu}} \bar{\mathcal{E}}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2} \\ &= \frac{4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} \mathcal{I}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3), \quad f = \pm.\end{aligned}\quad (146)$$

Here the index f labels the two nonlocal branches and is distinct from the SK index d of the box diagram. The loop integral can then be written in the same form as the triangle result:

$$\begin{aligned}\mathcal{LOOP} &= \sum_{\lambda_1, \lambda_2=L, \pm} \left[D^{(\lambda_1)}(k_1, \tau_1, \tau_2) \right]_{ab} \left[D^{(\lambda_2)}(k_3, \tau_3, \tau_4) \right]_{cd} \\ &\quad \times \sum_{f=\pm} (\tau_1 \tau_2 \tau_3 \tau_4)^{1/2+if\tilde{\nu}} \mathcal{B}_f^{\lambda_1\lambda_2}(k_1, k_3),\end{aligned}\quad (147)$$

where

$$\begin{aligned}\mathcal{B}_f^{\lambda_1\lambda_2}(k_1, k_3) &= \frac{\Gamma^4(-if\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}} \sum_{\sigma_1, \sigma_2=L, \pm} \mathcal{I}_{\sigma_1\sigma_2}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3) \\ &= \frac{\Gamma^4(-if\tilde{\nu})}{(4\pi)^{7/2}} 4^{-2if\tilde{\nu}} k_s^{3+4if\tilde{\nu}} \mathcal{I}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3)\end{aligned}\quad (148)$$

We then evaluate the contributions from the left and right subgraphs. Both are identical to the left-subgraph contribution in the triangle calculation:

$$\mathcal{T}_f^{(L), \lambda_1} = (ig^2 h_0)^2 \frac{k_{12}^{-2if\tilde{\nu}}}{4k_1^3 k_2^3} \cdot \mathcal{P}_f^{(2), \lambda_1}(\tilde{\nu}) \quad (149)$$

$$\mathcal{T}_f^{(R), \lambda_2} = (ig^2 h_0)^2 \frac{k_{34}^{-2if\tilde{\nu}}}{4k_3^3 k_4^3} \cdot \mathcal{P}_f^{(2), \lambda_2}(\tilde{\nu}) \quad (150)$$

Thus, the contribution of the box diagram is:

$$\langle \delta h^4 \rangle^{\text{Box}} = \sum_{f=\pm} \sum_{\lambda_1, \lambda_2=L, \pm} \mathcal{T}_f^{(L), \lambda_1} \cdot \mathcal{B}_f^{\lambda_1\lambda_2} \cdot \mathcal{T}_f^{(R), \lambda_2} \quad (151)$$

$$= (ig^2 h_0)^4 \frac{k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12} k_{34}} \right)^{2i\tilde{\nu}} \times C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \text{c.c.} \quad (152)$$

where the magnitude is:

$$C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = \sum_{\lambda_1, \lambda_2=L, \pm} \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot [\mathcal{P}_+^{(2), \lambda_1}(\tilde{\nu}) \mathcal{P}_+^{(2), \lambda_2}(\tilde{\nu})] \cdot \mathcal{I}^{\lambda_1\lambda_2}(\theta_1, \theta_3, \phi_3)|_{f=+} \quad (153)$$

Since $\mathcal{P}_+^{(2), +}(\tilde{\nu}) = \mathcal{P}_+^{(2), -}(\tilde{\nu})$, we define:

$$\mathcal{I}^{LH} = \mathcal{I}^{L+} + \mathcal{I}^{L-} + \mathcal{I}^{+L} + \mathcal{I}^{-L} \quad (154)$$

$$\mathcal{I}^{HH} = \mathcal{I}^{++} + \mathcal{I}^{--} + \mathcal{I}^{+-} + \mathcal{I}^{-+} \quad (155)$$

The magnitude can then be written as:

$$C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = \frac{\Gamma^4(-i\tilde{\nu})}{(4\pi)^{7/2}} \cdot \left\{ [\mathcal{P}_+^{(2), L}]^2 \cdot \mathcal{I}^{LL} + [\mathcal{P}_+^{(2), L} \mathcal{P}_+^{(2), +}] \cdot \mathcal{I}^{LH} + [\mathcal{P}_+^{(2), +}]^2 \cdot \mathcal{I}^{HH} \right\} \quad (156)$$

Through a lengthy calculation, we obtain an analytic expression for C^{Box} , which we write as:

$$C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = A_1 + A_2(\cos^2 \theta_1 + \cos^2 \theta_3) + A_3(\cos^2 \theta_1 \cdot \cos^2 \theta_3) \\ + B \sin(2\theta_1) \sin(2\theta_3) \cos \phi_3 + D \sin^2 \theta_1 \sin^2 \theta_3 \cos 2\phi_3 \quad (157)$$

For completeness, the five coefficient functions are extracted from the full box coefficient by the same angular projections used in the accompanying notebook. Define

$$\mathcal{C}(\theta_1, \theta_3, \phi_3) \equiv C^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3), \quad (158)$$

$$\mathcal{A}(\theta_1, \theta_3) \equiv \frac{1}{4} \left[\mathcal{C}(\theta_1, \theta_3, 0) + \mathcal{C}(\theta_1, \theta_3, \pi) + 2\mathcal{C}\left(\theta_1, \theta_3, \frac{\pi}{2}\right) \right]. \quad (159)$$

Then the explicit reconstruction formulae are

$$A_1(\tilde{\nu}) = \mathcal{A}\left(\frac{\pi}{2}, \frac{\pi}{2}\right), \quad (160)$$

$$A_2(\tilde{\nu}) = \mathcal{A}\left(\frac{\pi}{2}, 0\right) - A_1(\tilde{\nu}), \quad (161)$$

$$A_3(\tilde{\nu}) = \mathcal{A}(0, 0) - A_1(\tilde{\nu}) - 2A_2(\tilde{\nu}), \quad (162)$$

$$B(\tilde{\nu}) = \frac{1}{2} \left[\mathcal{C}\left(\frac{\pi}{4}, \frac{\pi}{4}, 0\right) - \mathcal{C}\left(\frac{\pi}{4}, \frac{\pi}{4}, \pi\right) \right], \quad (163)$$

$$D(\tilde{\nu}) = \frac{1}{4} \left[\mathcal{C}\left(\frac{\pi}{2}, \frac{\pi}{2}, 0\right) + \mathcal{C}\left(\frac{\pi}{2}, \frac{\pi}{2}, \pi\right) - 2\mathcal{C}\left(\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}\right) \right]. \quad (164)$$

These projections remove the unwanted angular structures and determine all five functions directly from the analytic expression for C^{Box} above.

Some numerical results are shown below:

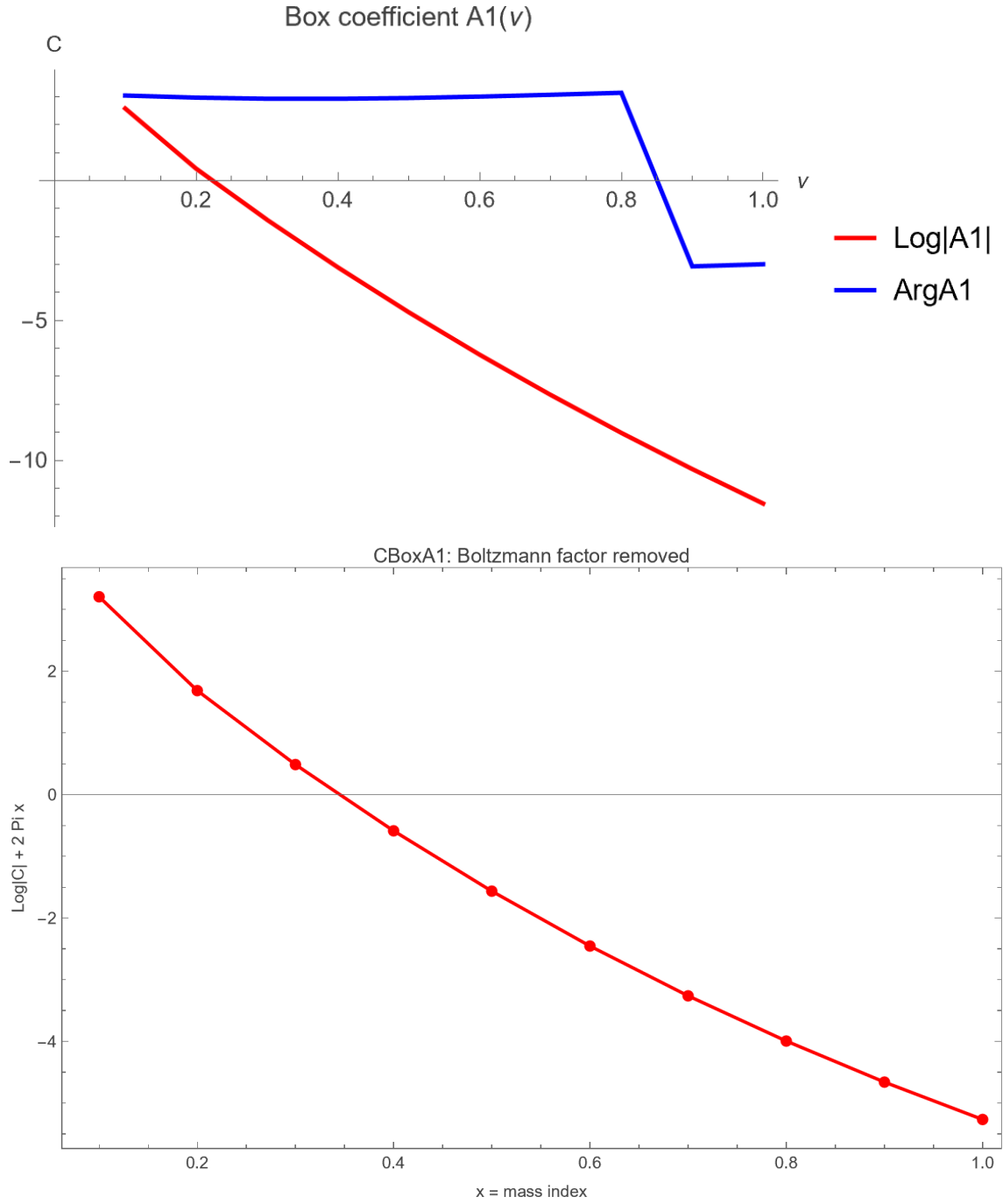


Figure 3: $A_1(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |A_1(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom).

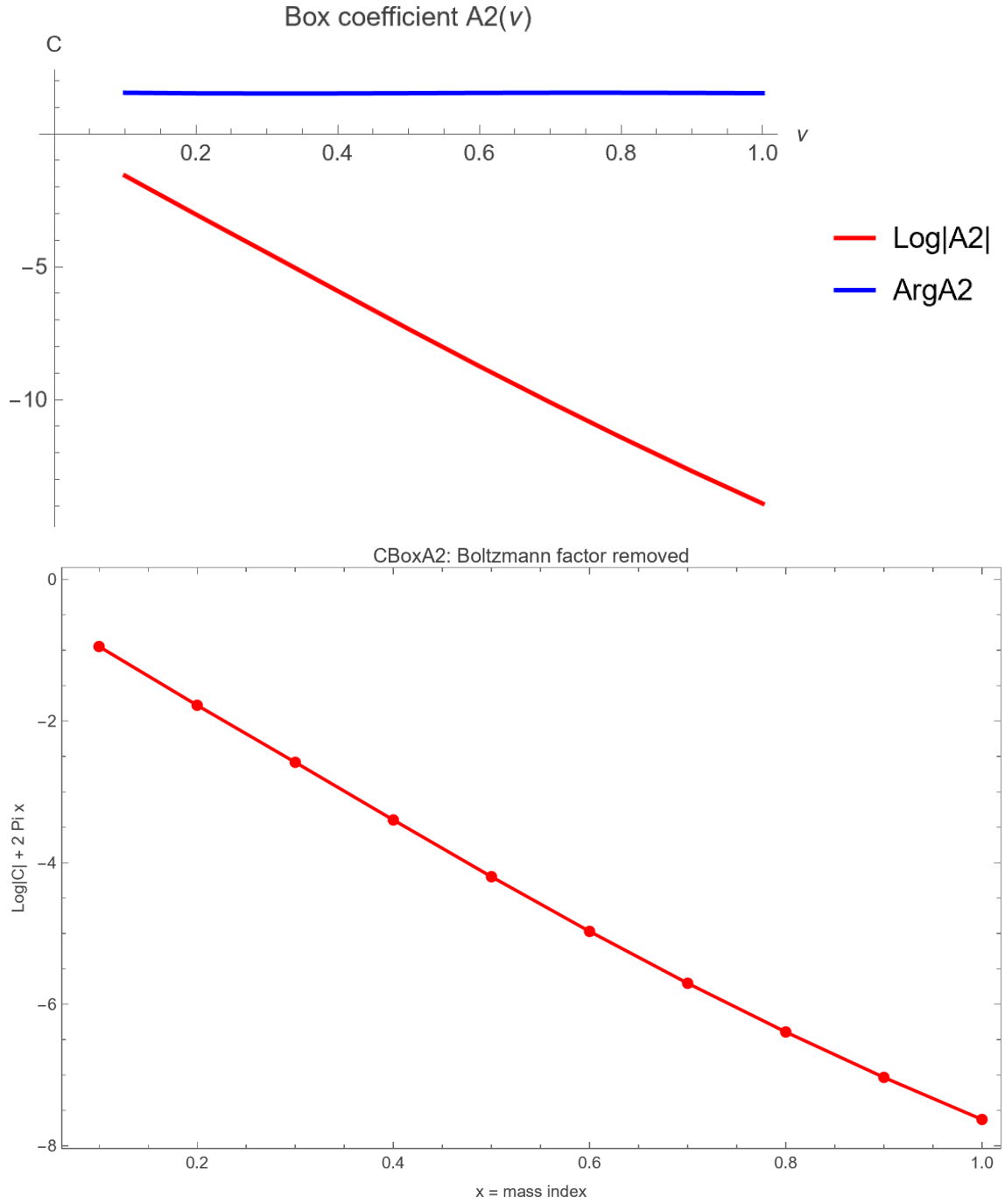


Figure 4: $A_2(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |A_2(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom).

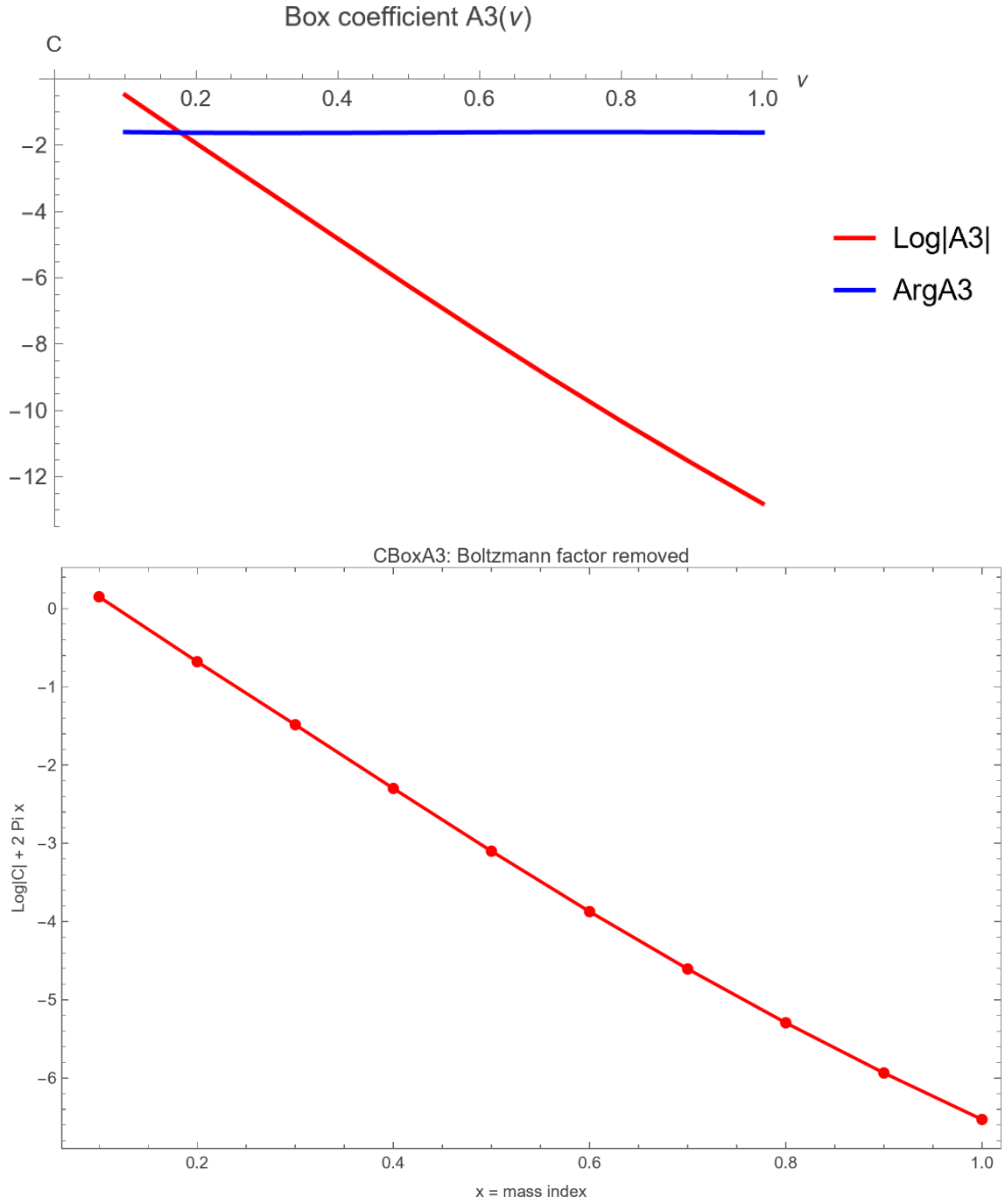


Figure 5: $A_3(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |A_3(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom).

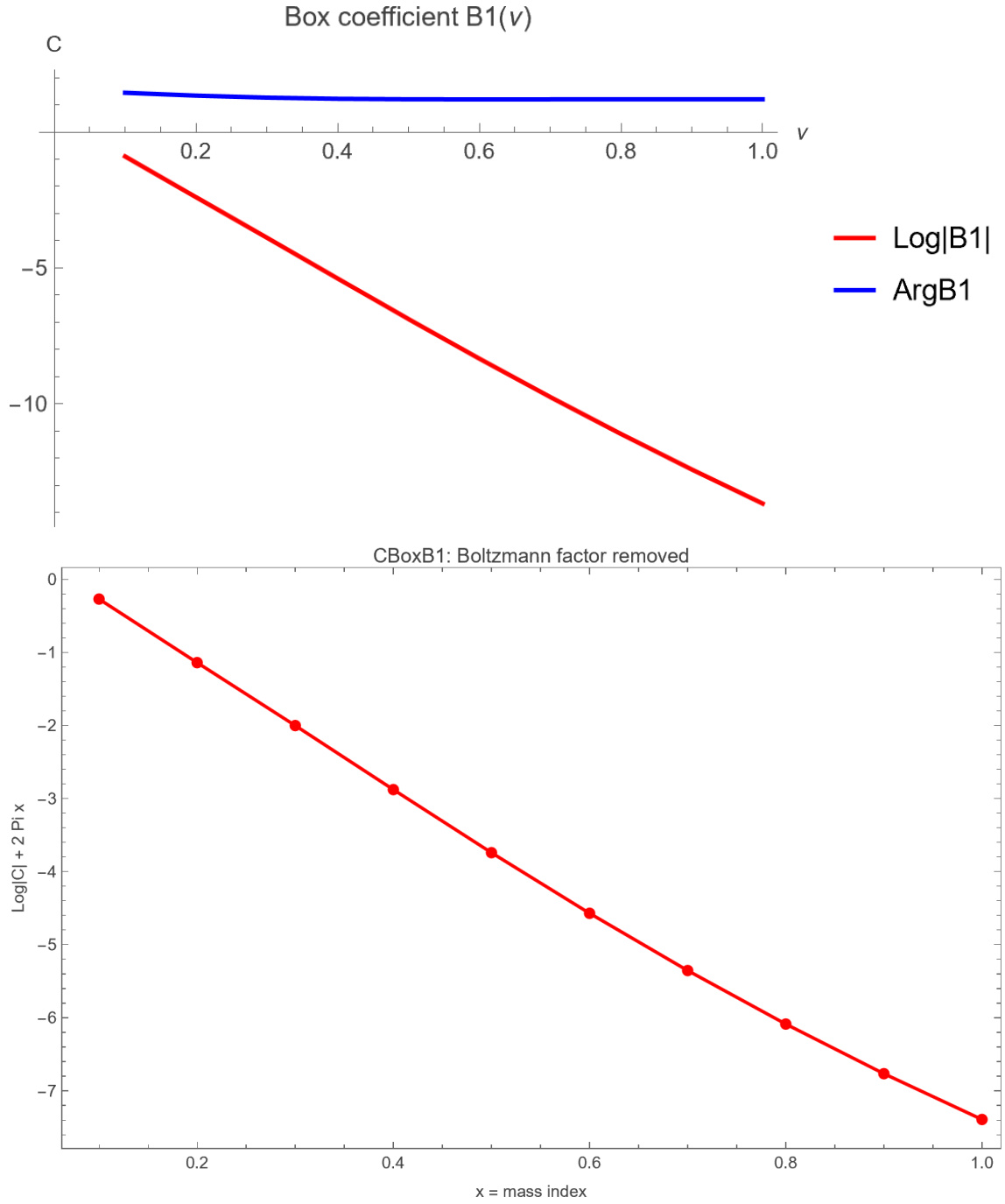


Figure 6: $B(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |B(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom).

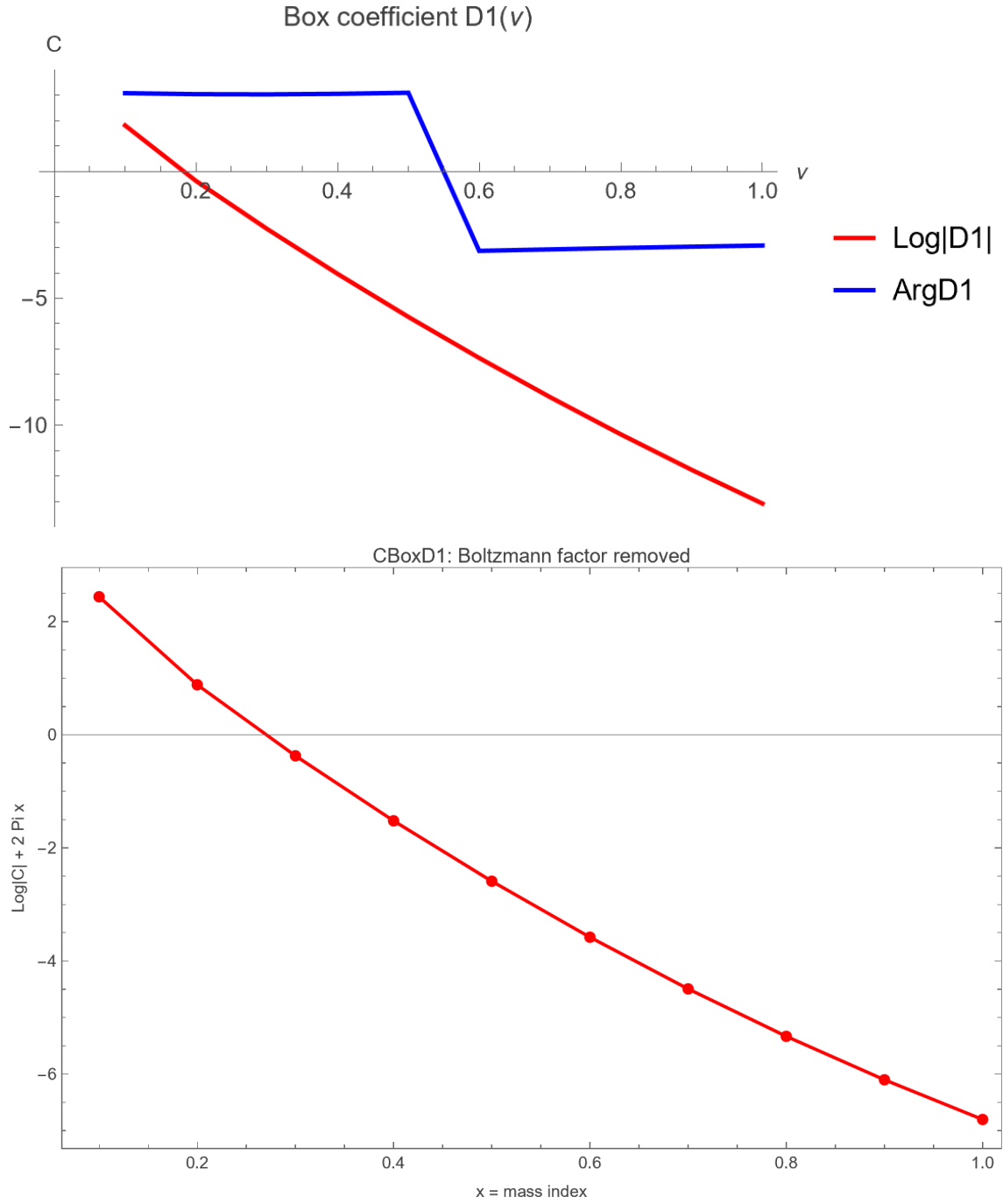


Figure 7: $D(\tilde{\nu})$ (top) and the rescaled log magnitude $\log |D(\tilde{\nu})| + 2\pi\tilde{\nu}$ (bottom).

A Closed Gamma-function forms for $\mathcal{J}_3$ and $\mathcal{J}_4$

For a general loop seed, write $\mathcal{J}_r(m, n) = f_r(a, b)$ with

$$a = x - \frac{m}{2}, \quad b = x + \frac{n}{2}, \quad x = -if\tilde{\nu}. \quad (165)$$

The identity $z = (K^2 - q^2 - k_s^2)/(2qk_s)$ gives the following finite Gamma-function closed forms:

$$\begin{aligned} f_3(a, b) = \frac{1}{8} & \left[f_0\left(a + \frac{3}{2}, b - 3\right) - 3f_0\left(a + \frac{3}{2}, b - 2\right) - 3f_0\left(a + \frac{1}{2}, b - 2\right) \right. \\ & + 3f_0\left(a + \frac{3}{2}, b - 1\right) + 6f_0\left(a + \frac{1}{2}, b - 1\right) + 3f_0\left(a - \frac{1}{2}, b - 1\right) \\ & \left. - f_0\left(a + \frac{3}{2}, b\right) - 3f_0\left(a + \frac{1}{2}, b\right) - 3f_0\left(a - \frac{1}{2}, b\right) - f_0\left(a - \frac{3}{2}, b\right) \right], \end{aligned} \quad (166)$$

$$\begin{aligned} f_4(a, b) = \frac{1}{16} & \left[f_0(a + 2, b - 4) - 4f_0(a + 2, b - 3) - 4f_0(a + 1, b - 3) \right. \\ & + 6f_0(a + 2, b - 2) + 12f_0(a + 1, b - 2) + 6f_0(a, b - 2) \\ & - 4f_0(a + 2, b - 1) - 12f_0(a + 1, b - 1) - 12f_0(a, b - 1) - 4f_0(a - 1, b - 1) \\ & \left. + f_0(a + 2, b) + 4f_0(a + 1, b) + 6f_0(a, b) + 4f_0(a - 1, b) + f_0(a - 2, b) \right]. \end{aligned} \quad (167)$$

Here f_0 is the Gamma ratio defined in the loop-seed section. Consequently, these equations are closed expressions in Gamma functions, not new independent seeds. More generally,

$$f_r(a, b) = \frac{1}{2^r} \sum_{p+q+t=r} \frac{r!}{p!q!t!} (-1)^{q+t} f_0\left(a + \frac{r}{2} - q, b - p\right), \quad (168)$$

which provides a direct check of both formulas.

References

- [1] Z. Qin and Z.-Z. Xianyu, “Helical Inflation Correlators: Partial Mellin-Barnes and Bootstrap Equations,” *Journal of High Energy Physics* **2023** no. 4, (Apr., 2023) 59. <http://arxiv.org/abs/2208.13790>. arXiv:2208.13790 [astro-ph, physics:hep-ph, physics:hep-th].
- [2] Z. Qin and Z.-Z. Xianyu, “Closed-Form Formulae for Inflation Correlators,” *Journal of High Energy Physics* **2023** no. 7, (July, 2023) 1. <http://arxiv.org/abs/2301.07047>. arXiv:2301.07047 [hep-th].